Table 1
Notation used by SC-FMA.

Symbol	Meaning	Domain
k	Number of reflective steps	$k \in \mathbb{N}$
τ	Reasoning trace	Text sequence with identified spans
$\mathcal{G}$	Reflection DAG	$(\mathcal{V}, \mathcal{E})$
c_i	Local CIU for step i	Protocol-dependent utility difference
n_i	Structural necessity for step i	$[0, 1]$
R_{ij}	Redundancy between steps i, j	$[0, 1]$ after PSD correction
b_i	Bottleneck indicator	$\{0, 1\}$
w_i	Calibrated supervision weight	$w_i \geq \varepsilon$
$\alpha, \beta, \gamma, \delta$	SCU trade-off weights	Nonnegative hyperparameters

Supplementary Material for Structurally-Calibrated Functional Attribution: A Methodology for Process Supervision Weighting in Reflective Reasoning

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ABSTRACT

This supplementary material provides notation details, proof sketches, additional experimental results, KBS pilot details, and reproducibility notes for the SC-FMA manuscript. It also records the validation boundary: GSM8K/HotpotQA replay and downstream filtering are failed or blocked provenance, while the Countries KG study is a fixture-level pilot over real KG structure.

1. Notation Reference

Table 1 summarizes the notation used in the main text. Vectors are column vectors, $\|\cdot\|_2$ denotes the Euclidean norm, and $\odot$ denotes elementwise multiplication.

2. Optimization Details

The SCU objective is:

$$L(\mathbf{w}) = \alpha\|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta\|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma\mathbf{w}^\top\mathbf{R}\mathbf{w} - \delta\sum_i b_i \log(w_i), \tag{1}$$

with $\mathbf{w}$ constrained to the simplex $\mathcal{W} = \{\mathbf{w} : w_i \geq \varepsilon, \sum_i w_i = 1\}$.

2.1. KKT conditions

For the constrained optimum $\mathbf{w}^*$, the Karush–Kuhn–Tucker conditions are:

$$\left.\frac{\partial L}{\partial w_i}\right|_{\mathbf{w}^*} - \mu_i^* - \lambda^* = 0, \tag{2}$$

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Table 2
Step importance ranking metrics stratified by trace size k. Values are mean Spearman ρ.

Method	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k \geq 7$
SC-FMA QP	0.582	0.601	0.615	0.622	0.634
SC-FMA Ridge	0.521	0.534	0.540	0.538	0.542
SC-FMA Projection	0.328	0.341	0.345	0.339	0.342
Raw CIU	0.471	0.485	0.490	0.482	0.486

$$\mu_i^* \geq 0, \tag{3}$$

$$\mu_i^*(w_i^* - \varepsilon) = 0, \tag{4}$$

$$w_i^* \geq \varepsilon, \tag{5}$$

$$\sum_i w_i^* = 1. \tag{6}$$

The derivative is:

$$\frac{\partial L}{\partial w_i} = 2\alpha(w_i - \tilde{c}_i) + 2\beta(w_i - \tilde{n}_i) + 2\gamma(\mathbf{R}\mathbf{w})_i - \delta b_i / w_i. \tag{7}$$

2.2. Variance-reduction intuition

If $\gamma = \delta = 0$, the unconstrained minimizer is:

$$\hat{\mathbf{w}} = \frac{\alpha\tilde{\mathbf{c}} + \beta\tilde{\mathbf{n}}}{\alpha + \beta}. \tag{8}$$

The calibrated vector therefore shrinks raw CIU toward structural necessity. Projection onto the simplex is non-expansive [2], so the constrained solution cannot amplify deviations in the same way direct raw-CIU weighting can. The main text reports the empirical consequence: SC-FMA QP improves ranking correlation over raw CIU while reducing sensitivity to noisy local utility.

3. Additional Experimental Results

3.1. Per-size ranking results

The improvement of SC-FMA QP over raw CIU grows with trace size, consistent with the mechanism that redundancy and bottleneck structure become more informative in larger reflection DAGs.

3.2. Structural diagnostic figures

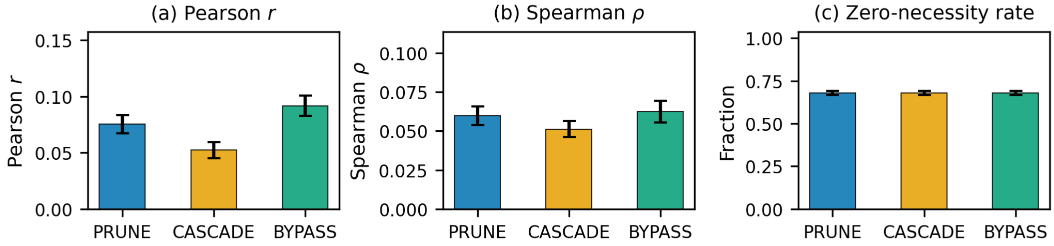

Figure 1: Structural diagnostic summary by perturbation mode. PRUNE, CASCADE, and BYPASS produce consistently weak CIU-necessity alignment.

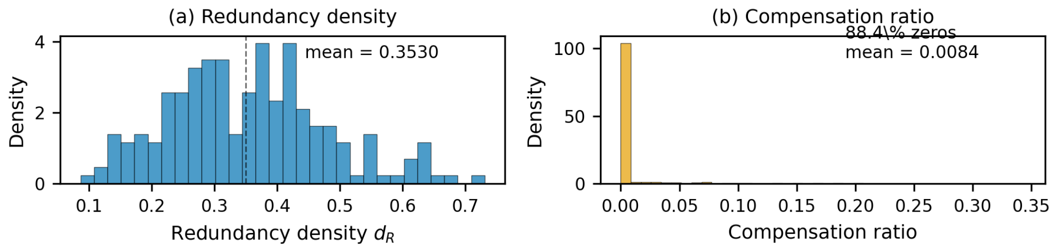

Figure 2: Redundancy and compensation diagnostics. Compensation is concentrated near zero, showing that removing a node rarely increases downstream necessity enough to replace it.

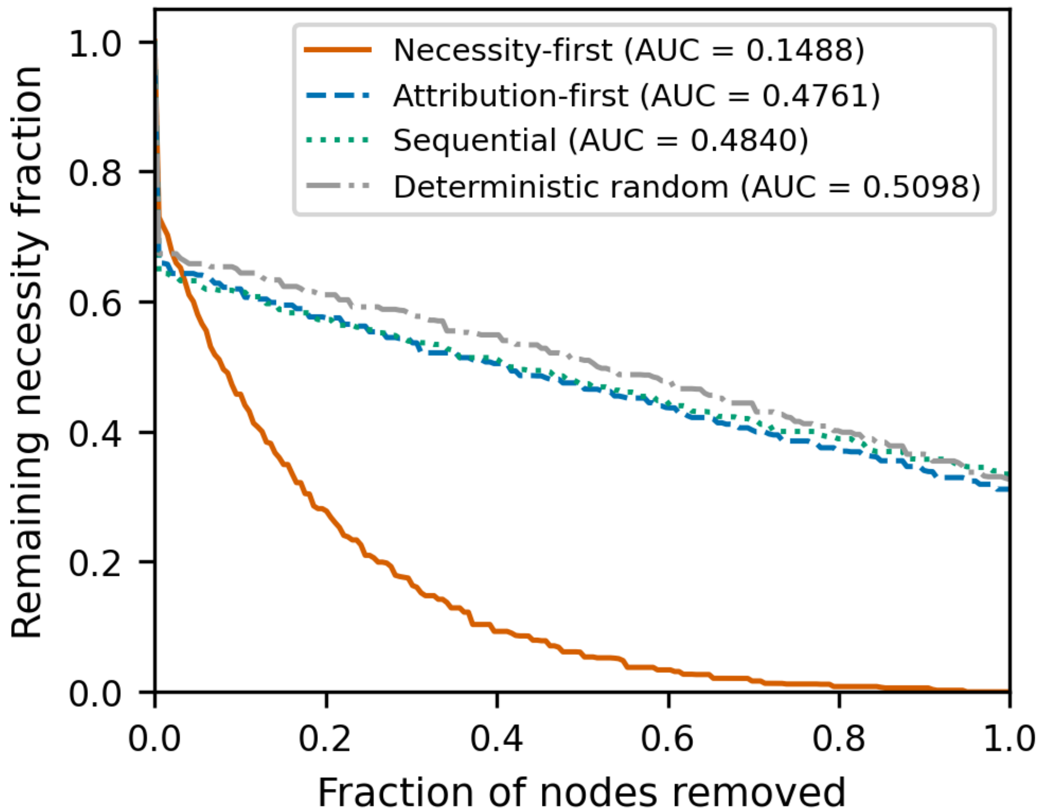

Figure 3: Resilience curves under ordered node removal. Necessity-first removal degrades structural support faster than attribution-first or random removal.

Table 3
KBS evidence boundary.

Evidence item	Status	Claim supported
Synthetic SC-FMA benchmark	Completed	Controlled calibration mechanism
PRM800K locked labels	Completed with overlap limits	Real step-label ranking context
Countries KG pilot	Pilot	Ontology-aware edge construction feasibility
GSM8K/HotpotQA replay	Failed/blocked	Provenance only
Downstream filtering	Failed/blocked	No performance-improvement claim
Rule-engine deployment	Not evaluated	Future work only

4. KBS Pilot Details

The Countries KG pilot is a small deterministic pilot over real KG structure [1]. It uses 25 country entities, 5 region entities, and 189 triples. The graph layer adds ontology-derived edges from `locatedIn` and `neighborOf` relations, then compares structural diagnostics with the baseline TF-IDF reflection DAG. The pilot result records `validated_kbs_workflow=false`: it checks ontology-aware edge construction feasibility, not deployed KBS correctness.

5. Reproducibility Notes

All reported synthetic experiments use deterministic seeds, fixed graph-construction parameters, and frozen hyperparameters after validation-set tuning. The source package includes both the manuscript and supplementary LaTeX sources, bibliography, CAS style files, and all figures used in the submission. Build artifacts are intentionally excluded from the upload source zip.

6. Submission Statements

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Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work.

Data Availability

Data will be made available on request.

References

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[2] Wang, W., Carreira-Perpiñán, M.Á., 2013. Projection onto the probability simplex: An ℓ_1-ball approach. arXiv preprint arXiv:1309.1541 URL: https://arxiv.org/abs/1309.1541.